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JEE MAIN PHYSICS DPP: COMPREHENSIVE PRACTICE MODULE

Topic: Practice Problems on Units, Errors & Dimensional Analysis | Time: 60 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** If speed of light c , acceleration due to gravity g , and pressure P are taken as fundamental physical quantities, the dimensional formula of gravitational constant G is:
- $[c^2 g^0 P^{-1}]$
 - $[c^0 g^2 P^{-1}]$
 - $[c^2 g^2 P^{-1}]$
 - $[c^1 g^1 P^{-2}]$
- Q2.** In an experiment to measure the density of a wire, a student records: mass $m = (0.3 \pm 0.003)$ g, radius $r = (0.5 \pm 0.005)$ mm, and length $l = (6.0 \pm 0.06)$ cm. The maximum percentage error in the calculated density is:
- 4%
 - 3%
 - 2%
 - 1%
- Q3.** The pitch of a screw gauge is 0.5 mm and its circular scale has 50 divisions. When measuring the diameter of a ball, the main scale reads 2.5 mm and the 20th circular division aligns with the reference line. If the zero error is +0.04 mm, the corrected diameter is:
- 2.66 mm
 - 2.70 mm
 - 2.74 mm
 - 2.64 mm
- Q4.** Which of the following expressions does **not** have the dimensions of time?
- RC
 - $\frac{L}{R}$
 - $\sqrt{LC}$
 - $\frac{R}{L}$
- Q5.** The position of a particle at time t is given by $x(t) = \left(\frac{v_0}{\alpha}\right)(1 - e^{-\alpha t})$, where v_0 is a constant velocity. The dimensional formula of α is:
- $[M^0 L^0 T^{-1}]$
 - $[M^0 L^1 T^{-1}]$
 - $[M^0 L^0 T^1]$
 - $[M^0 L^{-1} T^0]$
- Q6.** The value of resistance is $R = (10.0 \pm 0.1) \Omega$ and current through it is $I = (2.00 \pm 0.02)$ A. The thermal power dissipated ($P = I^2 R$) is:
- (40.0 ± 1.2) W
 - (40.0 ± 0.4) W
 - (40.0 ± 0.8) W
 - (40.0 ± 1.6) W
- Q7.** If 19 divisions of the main scale of a vernier caliper coincide with 20 divisions of the vernier scale, and 1 MSD = 1 mm, then the least count of the caliper is:
- 0.05 mm
 - 0.01 mm
 - 0.1 mm
 - 0.02 mm
- Q8.** In the dimensional formula $[M^a L^b T^c]$ of magnetic permeability μ_0 , the values of a, b , and c are:
- $a = 1, b = 1, c = -2$
 - $a = 1, b = 2, c = -2$
 - $a = 1, b = 1, c = -1$
 - $a = 1, b = 0, c = -2$
- Q9.** In a meter bridge experiment, unknown resistance $X = \frac{l}{100 - l} R$. If $R = 10.0 \Omega$ is known exactly and $l = (40.0 \pm 0.1)$ cm on a 100 cm wire, the fractional error $\Delta X/X$ is:
- $\frac{0.1}{24}$
 - $\frac{0.1}{40}$
 - $\frac{0.1}{60}$
 - $\frac{0.1}{100}$
- Q10.** A dimensionless quantity:
- Never has a unit
 - Always has a unit
 - May have a unit
 - Does not exist in physics
- Q11.** The capacitance C of a parallel plate capacitor is $C = \frac{\epsilon_0 A}{d}$. If percentage errors in plate area A and separation d are 1% and 2% respectively, the percentage error in C is:
- 3%
 - 1%
 - 2%
 - 0.5%

Q12. The velocity of sound in a medium depends on elasticity E and density ρ as $v = kE^a\rho^b$. The values of a and b are:

- (a) $a = 1/2, b = -1/2$
- (b) $a = 1/2, b = 1/2$
- (c) $a = -1/2, b = 1/2$
- (d) $a = 1, b = -1/2$

Q13. Perform the calculation $12.9 \text{ g} - 7.06 \text{ g}$ and express the answer to appropriate significant figures:

- (a) 5.84 g
- (b) 5.8 g
- (c) 5.840 g
- (d) 5.9 g

Q14. If force F , area A , and density ρ are chosen as fundamental quantities, the dimensional formula for velocity v is:

- (a) $[F^{1/2}A^{-1/2}\rho^{-1/2}]$
- (b) $[F^{1/2}A^{1/2}\rho^{-1/2}]$
- (c) $[F^{1/2}A^{-1/4}\rho^{-1/2}]$
- (d) $[F^1A^{-1}\rho^{-1/2}]$

Q15. A physical quantity y is given by $y = \frac{a^2b^3}{c\sqrt{d}}$. If percentage errors in a, b, c , and d are 2%, 1%, 3%, and 4% respectively, the percentage error in y is:

- (a) 12%
- (b) 10%
- (c) 8%
- (d) 14%

Q16. A screw gauge has a pitch of 1 mm and 100 circular scale divisions. When used to measure the thickness of a plate, the main scale reads 3 mm and the 35th circular division coincides with the datum line. If it has a negative zero error of 0.05 mm, the thickness of the plate is:

- (a) 3.40 mm
- (b) 3.30 mm
- (c) 3.35 mm
- (d) 3.45 mm

Q17. The equation of a stationary wave is $y = 2A \sin\left(\frac{2\pi ct}{\lambda}\right) \cos\left(\frac{2\pi x}{\lambda}\right)$. Which of the following statements is **incorrect**?

- (a) The dimension of ct is $[L]$.
- (b) The dimension of x is $[L]$.
- (c) The dimension of λ is $[L]$.
- (d) The dimension of c/λ is $[LT^{-1}]$.

Q18. The density of a material in CGS system is $8 \text{ g}\cdot\text{cm}^{-3}$. In a unit system where the unit of length is 5 cm and the unit of mass is 20 g, the numerical value of density is:

- (a) 50
- (b) 40
- (c) 20
- (d) 100

Q19. The time period of a simple pendulum is $T = 2\pi\sqrt{L/g}$. If the measured length is $L = (20.0 \pm 0.1) \text{ cm}$ and time for 50 oscillations is $t = (40 \pm 1) \text{ s}$, the percentage error in g is:

- (a) 5.5%
- (b) 3.0%
- (c) 4.5%
- (d) 2.5%

Q20. The inductive reactance $X_L = \omega L$ and capacitive reactance $X_C = \frac{1}{\omega C}$ both have the dimensions of:

- (a) Resistance
- (b) Capacitance
- (c) Inductance
- (d) Electric potential

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(a)	11	(a)	16	(a)
2	(a)	7	(a)	12	(a)	17	(d)
3	(a)	8	(a)	13	(b)	18	(a)
4	(d)	9	(a)	14	(c)	19	(a)
5	(a)	10	(c)	15	(a)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

$$[c] = [LT^{-1}], [g] = [LT^{-2}], [P] = [ML^{-1}T^{-2}].$$

$$\text{We know } [G] = [M^{-1}L^3T^{-2}].$$

$$\text{Let } G \propto c^x g^y P^z.$$

$$[M^{-1}L^3T^{-2}] = [LT^{-1}]^x [LT^{-2}]^y [ML^{-1}T^{-2}]^z$$

$$M : z = -1$$

$$L : x + y - z = 3 \Rightarrow x + y + 1 = 3 \Rightarrow x + y = 2$$

$$T : -x - 2y - 2z = -2 \Rightarrow -x - 2y + 2 = -2 \Rightarrow$$

$$x + 2y = 4$$

$$\text{Solving gives } y = 2 \text{ and } x = 0.$$

$$\text{Thus, } [G] = [c^0 g^2 P^{-1}].$$

Sol 2. Correct Option: (a)

$$\text{Density } \rho = \frac{m}{\pi r^2 l} \Rightarrow \frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + 2 \frac{\Delta r}{r} + \frac{\Delta l}{l}.$$

$$\frac{\Delta m}{m} = \frac{0.003}{0.3} = 1\%$$

$$\frac{\Delta r}{r} = \frac{0.005}{0.5} = 1\% \Rightarrow 2 \frac{\Delta r}{r} = 2\%$$

$$\frac{\Delta l}{l} = \frac{0.06}{6.0} = 1\%$$

$$\% \text{ error in } \rho = 1\% + 2\% + 1\% = 4\%$$

Sol 3. Correct Option: (a)

$$LC = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}.$$

$$\text{Observed} = 2.5 + (20 \times 0.01) = 2.70 \text{ mm}.$$

$$\text{Correct} = 2.70 - (+0.04) = 2.66 \text{ mm}.$$

Sol 4. Correct Option: (d)

- $[RC] = [T]$ (capacitive time constant)
- $[L/R] = [T]$ (inductive time constant)
- $[\sqrt{LC}] = [T]$ (period of LC resonance)
- $[R/L] = [T^{-1}]$ (represents frequency, not time).

Sol 5. Correct Option: (a)

The argument of the exponential function must be dimensionless:

$$[\alpha t] = [M^0 L^0 T^0] \Rightarrow [\alpha] = \frac{1}{[t]} = [T^{-1}]$$

$$\text{Thus, } [\alpha] = [M^0 L^0 T^{-1}].$$

Sol 6. Correct Option: (a)

$$\text{Nominal power } P = I^2 R = (2.00)^2 \times 10.0 =$$

$$40.0 \text{ W}.$$

Fractional error:

$$\begin{aligned} \frac{\Delta P}{P} &= 2 \left(\frac{\Delta I}{I} \right) + \frac{\Delta R}{R} \\ &= 2 \left(\frac{0.02}{2.00} \right) + \frac{0.1}{10.0} = 0.03 \end{aligned}$$

$$\text{Absolute error: } \Delta P = 0.03 \times 40.0 \text{ W} = 1.2 \text{ W}.$$

$$\text{Thus, } P = (40.0 \pm 1.2) \text{ W}.$$

Sol 7. Correct Option: (a)

$$20 \text{ VSD} = 19 \text{ MSD} \Rightarrow 1 \text{ VSD} = 0.95 \text{ mm}.$$

$$LC = 1 \text{ mm} - 0.95 \text{ mm} = 0.05 \text{ mm}.$$

Sol 8. Correct Option: (a)

$$\text{From force between parallel conductors: } \frac{F}{L} =$$

$$\frac{\mu_0 I^2}{2\pi d} \Rightarrow \mu_0 = \frac{2\pi d F}{I^2 L}.$$

$$[\mu_0] = \frac{[L][MLT^{-2}]}{[A^2][L]} = [M^1 L^1 T^{-2} A^{-2}]$$

$$\text{Therefore, } a = 1, b = 1, c = -2.$$

Sol 9. Correct Option: (a)

$$X = \frac{l}{100-l} R \Rightarrow \ln X = \ln l - \ln(100-l) + \ln R.$$

Differentiating for maximum error:

$$\frac{\Delta X}{X} = \frac{\Delta l}{l} + \frac{\Delta l}{100-l}$$

$$= 0.1 \left(\frac{1}{40.0} + \frac{1}{60.0} \right) = \frac{0.1}{24}$$

Sol 10. Correct Option: (c)

A dimensionless quantity can possess a unit (for example, plane angle is dimensionless with unit radian, solid angle with unit steradian).

Sol 11. Correct Option: (a)

$$C = \frac{\varepsilon_0 A}{d} \Rightarrow \frac{\Delta C}{C} = \frac{\Delta A}{A} + \frac{\Delta d}{d}.$$

$$\% \text{ error in } C = 1\% + 2\% = 3\%$$

Sol 12. Correct Option: (a)

$$[v] = [LT^{-1}], [E] = [ML^{-1}T^{-2}], [\rho] = [ML^{-3}].$$

$$[LT^{-1}] = [ML^{-1}T^{-2}]^a [ML^{-3}]^b$$

$$M : a + b = 0 \implies b = -a$$

$$T : -2a = -1 \implies a = 1/2 \implies b = -1/2.$$

$$\text{Formula: } v = \sqrt{E/\rho}.$$

Sol 13. Correct Option: (b)

$$12.9 - 7.06 = 5.84.$$

In subtraction, the result is limited by the least number of decimal places (12.9 has 1 decimal place).

Rounding 5.84 to 1 decimal place yields **5.8 g**.

Sol 14. Correct Option: (c)

Let $v \propto F^x A^y \rho^z$.

$$[LT^{-1}] = [MLT^{-2}]^x [L^2]^y [ML^{-3}]^z$$

$$M : x + z = 0 \implies z = -x$$

$$T : -2x = -1 \implies x = 1/2 \implies z = -1/2$$

$$L : x + 2y - 3z = 1 \implies 1/2 + 2y - 3(-1/2) = 1$$

$$1 \implies 2 + 2y = 1 \implies y = -1/4$$

$$\text{Thus, } [v] = [F^{1/2} A^{-1/4} \rho^{-1/2}].$$

Sol 15. Correct Option: (a)

$$y = \frac{a^2 b^3}{cd^{1/2}} \implies \frac{\Delta y}{y} = 2 \frac{\Delta a}{a} + 3 \frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{1}{2} \frac{\Delta d}{d}.$$

$$\% \text{ error} = 2(2\%) + 3(1\%) + 3\% + \frac{1}{2}(4\%)$$

$$= 4\% + 3\% + 3\% + 2\% = 12\%$$

Sol 16. Correct Option: (a)

$$LC = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}.$$

$$\text{Observed} = 3 \text{ mm} + (35 \times 0.01 \text{ mm}) = 3.35 \text{ mm}.$$

$$\text{Correct Thickness} = \text{Observed} - \text{Zero Error} = 3.35 - (-0.05) = 3.40 \text{ mm}.$$

Sol 17. Correct Option: (d)

$$\bullet [2\pi ct/\lambda] = 1 \implies [ct] = [\lambda] = [L].$$

$$\bullet \text{The speed of wave is } c, \text{ so } [c] = [LT^{-1}].$$

$$\bullet [c/\lambda] = \frac{[LT^{-1}]}{[L]} = [T^{-1}] \text{ (represents frequency } \nu, \text{ not speed).}$$

Sol 18. Correct Option: (a)

Density has dimensions $[M^1 L^{-3}]$.

$$n_1 = 8, M_1 = 1 \text{ g}, L_1 = 1 \text{ cm}.$$

$$M_2 = 20 \text{ g}, L_2 = 5 \text{ cm}.$$

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^1 \left(\frac{L_1}{L_2} \right)^{-3} = 8 \left(\frac{1}{20} \right) \left(\frac{1}{5} \right)^{-3}$$

$$n_2 = \frac{8}{20} \times 125 = \frac{2}{5} \times 125 = 50$$

Sol 19. Correct Option: (a)

$$T = t/n \implies \frac{\Delta T}{T} = \frac{\Delta t}{t} = \frac{1}{40} = 2.5\%.$$

$$\frac{\Delta L}{L} = \frac{0.1}{20.0} = 0.5\%.$$

$$\text{Since } g = \frac{4\pi^2 L}{T^2}:$$

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2 \frac{\Delta T}{T} = 0.5\% + 2(2.5\%) = 5.5\%$$

Sol 20. Correct Option: (a)

Reactances X_L and X_C represent the opposition to AC current and have unit Ohm (Ω), sharing identical dimensions with Resistance ($[M^1 L^2 T^{-3} A^{-2}]$).

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